

Homework II

Objective: Implement spatial convolution, color transformations, and histogram equalization from scratch using MATLAB, and compare your custom algorithms against MATLAB's built-in functions.

General Requirement for ALL Parts:

For **every task** in this assignment, you must:

1. Write your own algorithm from scratch using basic matrix operations (do not use built-in matlab functions).
2. Use built-in MATLAB functions (e.g., `imfilter`, `histeq`) to perform the exact same task.
3. Display the original image alongside the outputs of both your custom algorithm and the built-in function side-by-side.

Part 1: Custom Spatial Filter (Convolution)

- Setup: Load an RGB image, convert to double, and extract the R, G, and B matrices.
- Task: Define a **5x5 Averaging (Mean)** filter matrix. Apply your custom 2D convolution (handling edge **padding** mathematically) and a built-in filter to the Green channel only.
- Result: Reconstruct two new RGB images (converted back to uint8) representing the custom and built-in filtering results. Display them side-by-side with the original image.

Part 2: Filtering Luminance

- Setup: Using your original RGB matrices, calculate the Luminance matrix (Y) using the standard formula: $Y = 0.299R + 0.587G + 0.114B$.
- Task: Apply both your custom convolution function and a built-in filter to the Y matrix. Display all three Y matrices (original, custom filtered, built-in filtered) as grayscale images.
- Analysis: In your report, explain the mathematical difference between filtering the Y matrix versus filtering the R, G, and B matrices directly. Why does filtering Y keep the colors looking natural?

Part 3: Histogram Equalization

- Setup: Load a grayscale image.

- Task: Write a custom algorithm to perform histogram equalization from scratch (calculate the histogram, compute the Cumulative Distribution Function, and map the new intensities).
- Compare: Use MATLAB's built-in equalization function on the same original image.
- Result: Display the original, custom equalized, and built-in equalized images side-by-side. *(Bonus: Display the histograms of all three to visually verify your math).*

What to Submit

- Separate MATLAB scripts (.m) for Parts 1, 2, and 3. Ensure your custom functions are included in the scripts or the same folder.
- The RGB and grayscale image files used in your code.
- A short PDF containing your written analysis for Part 2.
- *Please compress all files into a single .zip folder.*